Reviewer #2:

In this review the authors made a metadata analysis of 797 published articles related to depuration rates of marine biotoxin in different species. From these they excluded 643 papers that were not related with marine species, not related with phycotoxins, or did not quantify or did not present data needed to determine depuration rates. They found that only 85 marine species have been studied for the phycotoxin depuration rates. Most of the studies have been done with Paralytic Shellfish Toxins in mollusks. They give many details about depuration rates according to species, toxin, tissues, and environmental conditions. By a Bayessian regression method they predict PST depuration rates for 102 unstudied bivalve species, and they also propose recommended monitoring frequencies according to the toxin depuration rates.

This analysis is quite complete and provides a clear understanding of the current state of knowledge with all phycotoxins and the species studied, in laboratories and in natural environments. The propose many actions that need to be taken in order for managers to improve their monitoring systems and they indicate the wide gaps there still regarding understanding the depuration rates of the different species, toxins and environmental conditions.

This article should be published, it will be of interest for many scientists.

> We are grateful for your close review of our manuscript, for this kind acknowledgement, and for your thoughtful and constructive feedback. We appreciate your comments and suggestions and detail how we addressed them in the indented blue text below.

I only have a few comments:

Introduction.
Line 40. presenting a risk in nearly all coastal waters.

> We adopted the suggested language: “More than 200 taxa of marine algae produce biotoxins that threaten human health (Lundholm et al., 2009), presenting risk in nearly all coastal waters (OBIS, 2025).”

Line 46. from a loss of commercial,…

> We adopted the suggested language: “As a result, managers often close fisheries and aquaculture to protect public health, which can cause significant socioeconomic impacts from a loss of commercial, recreational, and cultural activities (Moore et al., 2020).”

Line 47. Harmful Algal Blooms

> We opted not to spell out HABs here, since the acronym was defined in the first sentence of the introduction. We were careful to limit acronym use throughout the paper.

Specifically, we only use acronyms for HABs, PSTs (paralytic shellfish toxins), DSTs (diarrhetic shellfish toxins), and EEZs (exclusive economic zones).

Line 49. Human and animal populations.

We opted not to add “and animal” here given the focus of the paragraph and the paper on human health. However, we appreciate the sentiment of the suggestion, which is to highlight that understanding depuration rates is also useful for understanding how toxins move through the food web and threaten apex predators such as seabirds and marine mammals. We added the following underlined text to the discussion to highlight this:

> “Finally, the study of species that are not harvested by humans but represent important nodes in marine food webs is critical to understanding the trophic transfer of biotoxins (Holmes and Lewis, 2022; K. A. Lefebvre et al., 2002). Besides three unharvested copepod species (*Calanus finmarchicus* are harvested in Norway; (FAO, 2024)), we found that only two unharvested species (opossum shrimp, *Neomysis awatschensis* and Pacific mole crab, *Emerita analoga*) have had their depuration rates studied. Understanding the flow of biotoxins through marine food webs is important for anticipating and responding to seabirds and marine mammals suffering from toxicosis (Van Dolah et al., 2002).”

Lines 90-96. References for the one-compartment and two-compartment models are needed.

We added the following citations and references for one- and two-compartment models:

> “In a one-compartment model, biotoxins are assumed to accumulate within a single “compartment”, which can represent a single tissue, a group of tissues, or even a whole organism, and are assumed to be exponentially lost to the environment at a single rate (Silvert and Rao, 1992). For example, a toxin might accumulate in the digestive system following ingestion and then be quickly eliminated through egestion. In a two-compartment model, biotoxins are assumed to accumulate in an initial compartment (e.g., the digestive system) with some transference to a second compartment (e.g., muscle) with a slower exponential depuration rate (Silvert and Cembella, 1995).”

Silvert, W., Rao, D.V.S., 1992. Dynamic Model of the Flux of Domoic Acid, a Neurotoxin, through a Mytilus edulis Population. Can. J. Fish. Aquat. Sci. 49, 400–405. https://doi.org/10.1139/f92-045

Silvert, W.L., Cembella, A.D., 1995. Dynamic modelling of phycotoxin kinetics in the blue mussel, Mytilus edulis, with implications for other marine invertebrates. Can. J. Fish. Aquat. Sci. 52, 521–531. https://doi.org/10.1139/f95-053

Line 124. Fernández

Thank you for catching this oversight. We added the missing accent in the citation and reference.

Line 132. García

Thank you for catching this oversight. We added the missing accent in the citation and reference.

Results

Line 369. Laboratory

We spelled out “laboratory” here: “Five studies directly compared depuration rates in the laboratory, where organisms are fed a controlled non-toxic diet, versus the field, where organisms experience a natural diet that may not be completely toxin-free (Fig. 3G).”

We also replaced all other instances of “lab” with “laboratory” in the manuscript.

Line 401. Satiated?

We replaced “less hungry” with “more satiated”.

Line 431, eliminate -coma- after et al.

Thanks for spotting this typo. We deleted the comma: “Bogan et al. (2007) found…”

Discussion

Line 623. Starved

Thanks for spotting this typo. We replaced “staved” with “starved”.

Line 693. Although bivalves have been the most studied organisms,

We adopted the suggested language: “Although bivalves have been the most studied organisms, depuration rates have still not been quantified for most bivalve species harvested in commercial fisheries and aquaculture.”

Line 718. McGuire

Thank you for catching this oversight. We corrected the capitalization in the citation and reference.

References

All scientific names should be in italics

We reviewed the references and italicized all scientific names.

Figure 1. The explanation of the circles is confusing, the standard monitoring program includes the green AND orange circles, not only the green circles. This should be indicated.

This is a good point. We changed the label for the green points to read “Unneeded” rather than “Standard” since these signify the points from the standard monitoring schedule that are unneeded. The caption is consistent with this change.

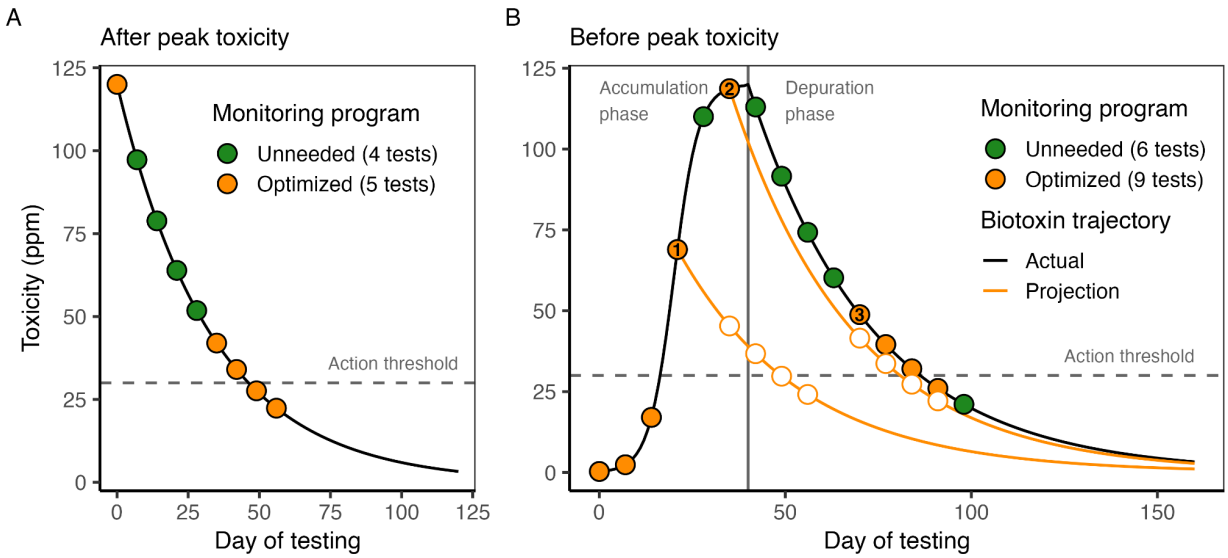


Figure 1. *An illustration of the value of biotoxin depuration rates in efficiently monitoring and managing biotoxin risk both* ***(A)*** *after and* ***(B)*** *before peak toxicity is attained. In both panels, the black curve shows a hypothetical accumulation (B only) and depuration trajectory for a marine seafood species that depurates at 0.03 day^{-1}. The points indicate weekly sampling that ceases once toxicity falls below the management action threshold (horizontal dashed line) in two consecutive weeks, a common requirement for opening a fishery closed due to biotoxin risk. In* ***(A)****, standard weekly sampling would require nine tests (all points), but pausing testing until two weeks before toxicity is projected to fall below the action threshold would require only five tests (orange points only). In* ***(B)****, scheduling tests based on three projections (marked by numbers) eliminates the need for six tests. The white circles indicate the projected testing schedule that is updated when toxicity is found to be higher than expected.*

Reviewer #3:

This submission is of highly valuable interest to the readers of Harmful Algae and should definitely be published after performing several modifications/improvements.

This manuscript represents a tremendous amount of analysis work, which is a high value for management for fisheries and aquaculture in regions subjected to HAB, or in regions where news HABs could occur. I would like to congratulate the authors for such a study!

The thorough analysis of the literature clearly demonstrate the effect of the environment (via food, temperature), of the exposure intensity (and its complexity), the physiological parameters intrinsic to the toxin accumulator (size, age and growth), as well as the differences in body parts or field versus lab studies on the accumulation and depuration rates.

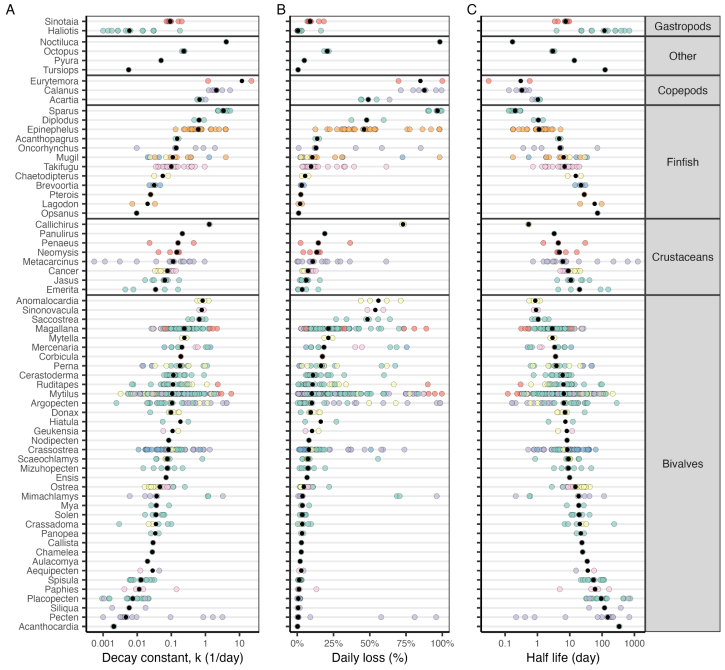
> We are grateful for your close review of our manuscript, for this kind acknowledgement, and for your thoughtful and constructive feedback. We agree with all of your comments and suggestions and detail how we addressed them in the indented blue text below.

The potential difference in depuration rates between different toxins for a same species should also be developed. The paper would gain from not pooling all toxins, but showing the specificity of each toxin (see comments below).

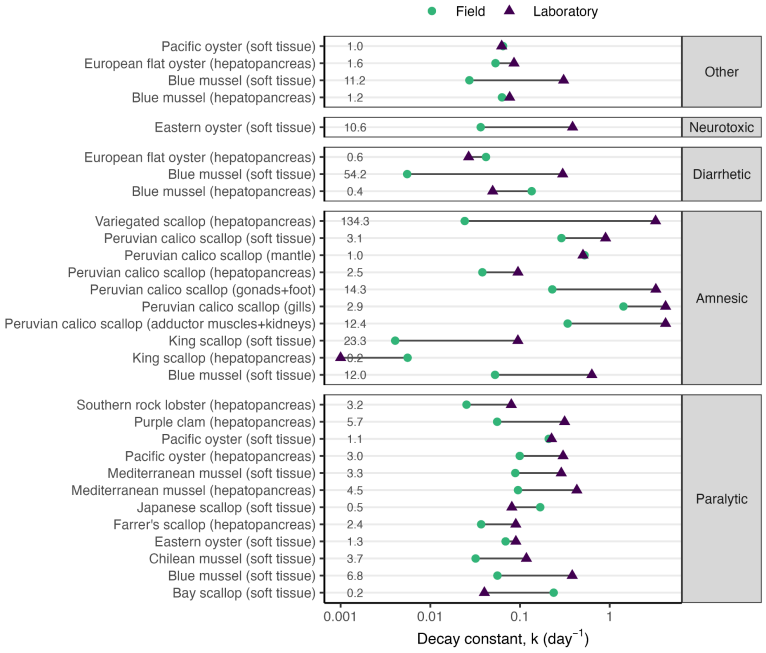
> This is a very good suggestion and we took several actions to better highlight and discuss variation in depuration rates between toxins in the same species.
>
> First, we changed **Figure 4** to show variation in depuration rates by biotoxin. We removed the boxplots and plot the raw data colored by biotoxin.
>
> Second, we moved **Figure S6**, which shows average field versus. laboratory depuration rates broken down by toxin, species, and tissue, to the main text. It is now **Figure 5**.
>
> Third, we expanded the description of this variation in the results by adding the following underlined text:
>
>> "Depuration rates varied from 22.3 day^{-1} (0.03 day half-life) for microcystin from whole *Eurytemora affinis* copepods (Karjalainen et al., 2006) to 0.0005 day^{-1} (1269 day half-life) for domoic acid from the gills of Dungeness crab (*Metacarcinus magister*) (Schultz et al., 2013) (**Fig. 4**). <u>In general, depuration rates varied more across genera than within genera (**Fig. 4**), motivating the taxonomic regression analysis described in *Section 2.3*. Even so, considerable variation in depuration rates within genera results from differences in depuration rates between species, tissues, toxins, and environmental conditions (**Figs. 4 & 5**). Most notably, depuration rates were generally much faster in the laboratory than in the field (**Fig. 5**). Furthermore, many genera show differences in</u>

depuration rates among biotoxins (**Figs. 4**). For example, *Ostrea* oysters depurate DSTs slower than PSTs and pectenotoxins and *Cancer* crabs also depurate DST slower than pectenotoxins. *Crassostrea* oysters depurate brevetoxin slower than domoic acid and *Argopecten* scallops depurate PSTs slower than domoic acid (**Fig. 4**)."

Figure 4. Revised to show variation by biotoxin.



New **Figure 5** (former **Figure S6**).



The second part of the study focusses on priority species to be analysed for depuration rates, to better predict overall depuration rates of most exploited species. This analysis is of tremendous importance and interest. However, it is still relatively complicated to follow the different steps (as described below). This section could gain in interest if details were provided as to how the number of exploited and total species have been determined and the choice of species to prioritize to more thoroughly explained?

We took several actions to address this comment.

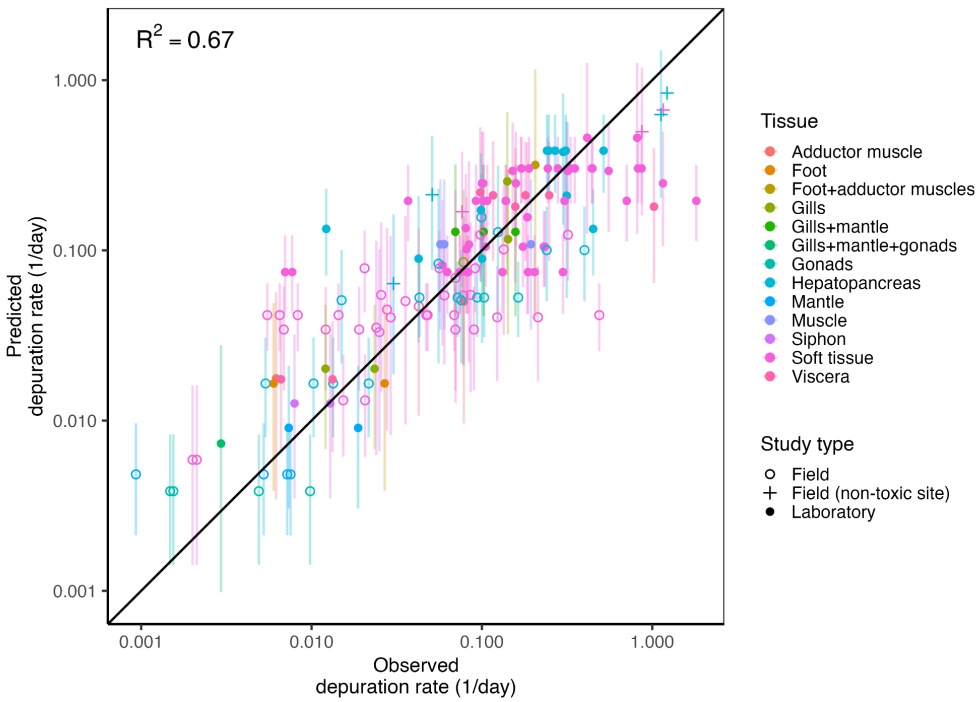
First, we significantly revised the methods section describing the prioritization process. Based on the comments below, the reviewer understood the methods for identifying priority species (highly produced species that occur in areas of HAB risk but that haven't had their depuration rates studied), but was confused by the additional contextual metrics that we assembled to further support prioritization. Thus, our edits focussed on clarifying the derivation of those contextual metrics and on clarifying **Table S4** where those metrics are described. This is described in the detailed comments below.

Second, we simplified the way these results are described, which greatly increases clarity and also improves readability by making the results more concise. This is also described in the detailed comments below.

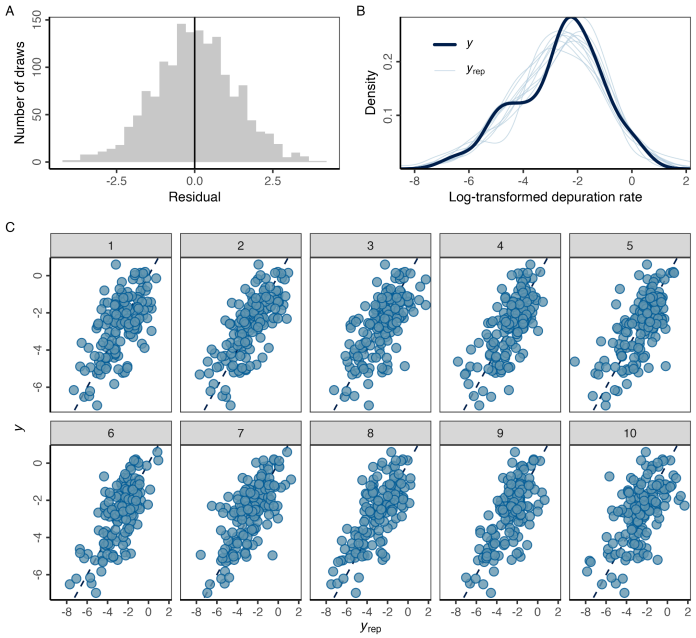
The last section, which predicts specific depuration rate for PST for a large list of species is one of the most important information of the paper: it will be extremely useful! However, the appropriation of the data the way it is currently presented is very difficult and the manuscript would highly benefit from improving this section. Several comments are made below. A major one would be that it is difficult for the reader to compare the prediction to the real values of depuration rates when they exist. This could be added to the table and figure showing the prediction. It would also give some more credibility to all the predicted values.

We took several actions to address this important comment.

First, we added a new figure to the supplement (**Figure S8**) to illustrate the correlation between the observed and predicted PST depuration rates in bivalves. This figure explicitly shows how well the model fits the data. The Bayesian R^2, which was reported in the results, is also printed in the top-left corner of the plot.



Second, we expanded the caption for **Figure S8** to increase clarity for readers with less familiarity with Bayesian modelling. This figure shows the posterior predictive check plots that are commonly used to examine the fit of Bayesian models to data.



Specifically, we added or significantly revised the following underlined text, trying to balance Bayesian reporting conventions with readability for non-Bayesian modelers.:

"**Figure S9.** Diagnostic plots for <u>the taxonomically nested random effects model (the best performing model)</u> showing **(A)** the distribution of the residuals in 10

posterior draws; **(B)** distribution of the response variable (log depuration rate) in the data (y) versus 10 posterior draws (y_{rep}); and **(C)** correlation between the response variable in the data (y) versus 10 posterior draws (y_{rep}). In **(C)**, the diagonal line represents the one-to-one line and each facet represents a set of posterior draws. The model was judged to pass these posterior predictive checks because the **(A)** residuals are normally distributed around zero; **(B)** the shape of the distributions of predicted depuration rates (y_{rep}) match the shape of the distribution of observed depuration rates (y); and **(C)** the predicted depuration rates (y_{rep}) are correlated to the observed depuration rates (y) without concerning patterns of bias."

Another comment regarding this section is that is is limited to PST : why? Is it because the lack of species-toxin combination with other toxins? Could it be do-able for ASP? If yes, it would be very interesting to propose the same results for other toxins. If not possible, it would be good to explain why this was only done for PST and what is missing to do it for other toxins. This could be added in the discussion.

Yes, we only attempted to model/predict bivalve PST depuration rates based on our *a priori* expectation that this would be the only clade-syndrome with sufficient data to support a model with good predictive skill. Our database includes 312 PST depuration rates for 36 bivalve species (12 families). In comparison, bivalve DST depuration rates, the clade-syndrome with the second best representation in the database, are described by only 98 rates from 15 species (5 families). This is a third of the PST data and includes much less taxonomic diversity to support good predictions. A large sample size is especially critical given that rates come from multiple tissue types. The database includes even less data for bivalve AST rates: 88 rates for 11 species (5 families).

We added the following underlined text to clarify this throughout the manuscript:

Final introductory paragraph: "Finally, we used a Bayesian regression model trained on our database of biotoxin depuration rates to predict PST depuration rates for unstudied bivalve species, the only clade-syndrome with sufficient data to support such an analysis."

Methods: "We compared the ability for one generalized linear model and three mixed effects models to predict paralytic shellfish toxin (PST) depuration rates in marine bivalves. We focused on PST depuration rates in bivalves based on the *a priori* expectation that this was the only clade-syndrome combination with sufficient data (**Fig. 2**) to generate a model with good predictive skill. Our database includes 312 PST depuration rates from 36 bivalve species (12 families); in comparison, bivalve diarrhetic shellfish toxin depuration, the clade-syndrome with the next best representation in the database, is described by only 98 rates from 15 species (5 families)."

We note that the reasons for our focus on bivalve PSTs is also mentioned in the abstract:

> **Abstract:** "Finally, we used a Bayesian regression model to predict PST depuration rates for 102 unstudied bivalve species, the only clade-syndrome with sufficient training data."

Again, this submission is of great interest and represent very valuable work, it should definitely be published after addressing the several comments, especially to gain in clarity.

Thanks again for your close review of our manuscript, for this kind acknowledgement, and for your thoughtful and constructive feedback. The manuscript has greatly improved as a result of your comments and suggestions.

Specific comments:

Introduction: very well written and excellent state of the art. It places very well the research in the context and poses very relevant questions.

Thank you for this acknowledgement.

A general comment on this publication is that species names (generic or Latin) should always be the same and consistent across the different parts: tables, figures and text of the submission.

We have reviewed and edited the manuscript for consistency.

In the materials and methods section, which is very well described, only the description of the four models could be refined a little, as the authors attempt to explain the differences between the four models, but this could be described a little better. In addition, the terminology used for the names of the models is not the same as in the results section or in the table section. Using the same terminology would make it much easier to read. For example, taxonomic nested model becomes taxonomic nested random model in the table.

We revised the text, tables, and figures to consistently describe the four models as the:

1. "Fixed effects only" model
2. "Phylogenetic random effects" model
3. "Taxonomically nested random effects" model
4. "Species random effects" model

We appreciate you flagging this inconsistency, as this would have caused confusion.

Results
Line 315 is Figure 2 and not Figure 2A.
Lines 323 and 326 are Figure 2 and not Figure 2B.

Alternatively, the letters A and B should be added to Figure 2.

> We reduced “2A” and “2B” to just “2”. Thanks for spotting this inconsistency.

Figure S6 should be included in the publication as it contains a large amount of data that would allow comparisons of the estimates made for all depuration rates.

> We appreciate this encouragement and added **Figure S6** to the main text. It is now **Figure 5**. We also expanded **Figure 4** and the results section to offer improved description of differences in depuration rates between toxins within the same species. See our response to a related comment in the major comments for details.

Lines 351 and 352: add the fact that his work refers to Figure 3D.

> We added a reference to **Fig. 3D** here: “Four studies (3%) conducted experiments in which organisms that were fed toxic algae in the lab underwent depuration at non-toxic field sites to mimic depuration in a natural, but non-toxic, setting (**Fig. 3D**).”

Line 351: a word is missing: Four studies (3%) compare depuration of

> We agree that this sentence was confusingly worded. We entirely rewrote it as:
>
> > “Four studies (3%) conducted experiments in which organisms that were fed toxic algae in the lab underwent depuration at non-toxic field sites to mimic depuration in a natural, but non-toxic, setting (**Fig. 3D**).”

Figure 3 is very comprehensive and provides a great deal of information. Panels D, E and F could be moved to the supplementary material and Figure S6 could be moved to the main text.

> We appreciate this encouragement and moved **Figure S6** to the main text (now **Figure 5)**. We made it a standalone figure and retained all of the original panels in **Figure 3**.

Indeed, In panel D, the caption lacks clarification: there is no mention of field studies on the graph, which is found in the caption: is it possible to homogenise?

> We revised the caption to clarify that panel D only shows the diet treatments used in lab studies since all field studies used wild diets:
>
> > “Panel **D** shows the feeding treatment used during the depuration phase of laboratory studies (all field studies used wild diets).”

Panel the difference between 0 and mixture is also unclear and should be explained

Thanks for flagging this because neither “zero” nor “mixture” were well-explained. We took several actions to correct this.

First, we added the following text to the methods to explain a zero-compartment model:

> “If the depuration rate was modelled, we recorded the model type (i.e., a zero-, one-, two-, or three-compartment model) and the estimated rate. A zero-compartment model describes depuration as linear decay (slope) whereas the other models describe depuration as exponential decay (decay constant or half-life) from the corresponding number of compartments (Blanco, 2009).”

Second, we added the following text to the **Figure 3** caption to explain that “mixture” pertains to papers that use different models for different species or toxins:

> “Panel **E** shows the modeling method used to quantify depuration rates (i.e., zero, one-, two-, or three-compartment exponential decay models). One paper used different modelling methods for different toxins and was recorded as using a “mixture” of one- and two-compartment models.”

Third, we changed “none” to “no model” in the figure to disambiguate “none” from “zero”.

Figure S6 should be part of the manuscript (and not in the supplementary material). This data is important for the reader and should be an integral part of the manuscript, as it is the kind of information that readers will be looking for.

We appreciate this encouragement and added **Figure S6** to the main text. It is now **Figure 5**. We also expanded **Figure 4** and the results section to offer improved description of differences in depuration rates between toxins within the same species. See our response to a related comment in the major comments for details.

Line 394: add 'for fed versus starved individuals in six different studies' for

We reworded this as: “Faster depuration has been found for fed versus starved individuals in studies on: (i) PSTs in Pacific oysters…”

Line 426: change warming to warm waters.

We reworded this sentence as: “Finally, Braga et al. (2018) is the only paper to find slower depuration in warmer water, where Mediterranean mussels depurated PSTs faster at 19°C than at 24°C.”

Lines 429 to 437: the authors should specify the purification time for the studies listed. It is unclear whether this purification may be due to growth, given that the purification time for the studies is not specified.

We added the following underlined text to provide the purification time, discuss the role of body growth in these results, and add a paper that was erroneously missing:

> "The five papers that considered the impact of size or age on biotoxin depuration rates generally found that smaller/younger individuals depurate faster than larger/older individuals. Over a 1-year depuration period, Bogan et al., (2007) found that smaller king scallops (*Pecten maximus*) depurated domoic acid faster than larger scallops. Over a 14 day depuration period, Mafra et al., (2010) found that smaller Eastern oysters (*Crassostrea virginica*) depurated domoic acid faster than larger oysters but that size did not affect the domoic acid depuration rates of blue mussels (*Mytilus edulis*). Similarly, (Duinker et al., 2007a) found no difference in the okadaic acid depuration rates of 1- and 2-year-old blue mussels over a 75-day depuration period. However, over a 5-day depuration period, Novaczek et al. (1993) found that smaller blue mussels depurated domoic acid faster than larger mussels. Finally, over a 7-day depuration period, Min et al., (2018) show that cyanotoxin depuration began earlier for juvenile mysid crustaceans (*Neomysis awatschensis*) and thus progressed faster. Faster apparent depuration rates in younger individuals could simply be due to faster body growth at small sizes, as growth inherently reduces the concentration of a constant amount of toxin, especially for fast-growing species with slow depuration rates observed over long periods."

Line 450: the authors discuss the addition of N-acetylcysteine as a potential agent to accelerate the detoxification of domoic acid by scallops, yet a study by Vanmalderghem et al 2023 demonstrates the opposite.

Thank you for letting us know about this study, which was not discovered in the Web of Science search. We added the following sentence to highlight this result:

> "However, in a subsequent replication study, Vanmaldergem et al. (2023) found no impact of N-Acetylcysteine on the depuration of domoic acid from king scallops."

Vanmaldergem, J., García-Corona, J.L., Deléglise, M., Fabioux, C., Hegaret, H., 2023. Effect of the antioxidant N-acetylcysteine on the depuration of the amnesic shellfish poisoning toxin, domoic acid, in the digestive gland of the king scallop *Pecten maximus*. Aquat. Living Resour. 36, 14. https://doi.org/10.1051/alr/2023011

Lines 475 to 479: these data should be developed much further. In my opinion, this is the core of the paper and the important information provided by this study.

One problem with this paragraph is that the information is pooled by type of organism, but without distinguishing between toxins. This may not be the appropriate strategy. Separating by toxin would be more relevant.

However, emphasising the depuration rates and the difference between toxin and organism seems very important, and this paragraph deserves to be expanded.

> We took several actions to address this important comment. Briefly, we (1) expanded **Figure 4** to illustrate variability in depuration rates within species by biotoxin; (2) moved **Figure S6** to the main text (now **Figure 5**); and (3) expanded the description of these results in *Section 3.1.3*. See our response to a related comment in the major comments for more details on these actions.

This work is represented in Figure 4, which is very interesting/informative and is one of the key figures in the paper. However, again, it would have been interesting to separate the data by toxin or at least to use different colours for the different toxins. Indeed, some species do not respond in the same way in terms of purification from one toxin to another, as is the case with Pecten maximus, for example (slow purifier of domoic acid, but not PSTs), and this figure would allow this observation to be made. Is this feasible?

> We changed **Figure 4** to show variation by biotoxin. See our response in the major comments for how we addressed this important issue more broadly.

In some cases, the box plot shows very large differences between purification rates. Is it possible to discuss these differences?

> We changed **Figure 4** to show variation by biotoxin and added **Figure 5** to show variation by environment (lab vs. field). This helps to understand the source of variability. Species and tissue type also contribute to this variability but there are too many species and tissue types to plot them in a way that would make them individually distinguishable.

Section 3.2 on species to be analysed as a priority for purification studies is not very clear, although it is very interesting.

> We agree and refer you to our response to the same comment related to Line 486 a few comments below to see how to simplified these results to increase their clarity.

Figure 5: Is it possible to clarify in the figure caption that the species are listed according to the regions in which the different species of toxic algae bloom? This explains the differences in species production and their order of importance.

> This is explained by the underlined text in the figure caption, but the wrong supplemental figure was cited before (**Fig. S5** instead of **Fig. S4**). We hope that correcting this error will help the reader understand the existing caption text better:
>
> > "The twenty filter-feeding marine mollusc species with the greatest annual production (millions mt, 2014-2023) <u>within Exclusive Economic Zones exposed to</u>

harmful algal blooms causing each of the evaluated biotoxin syndromes (**Fig. S4**) and an indication of whether (orange) or not (green) the depuration rate of the causative biotoxin has been studied."

Figure 5A: Flipper cupped oyster and stripped Venus are listed but do not have the same names in Table S4 or in the text. Is it possible to standardise the names? Similarly, in Figure 5B, Japanese carpetshell is named Manila clam in the text, and these are just a few examples.

We reviewed the text, tables, and figures for consistency in common names.

- We ensured that *Magallana bilineata* is called "slipper cupped oyster" throughout the text.
- We ensured that *Chamelea gallina* is called "striped venus" throughout the text. It was called "warty venus" in a few places.
- We ensured that *Ruditapes philippinarum* is called "Manila clam" throughout the text, tables, and figures. It was called "Japanese carpet shell" in a few places.

Table S4 should be clarified: What do the last two columns represent and how was this information collected? It is not sufficiently clear, either in the text or in the table caption.

We took two actions to clarify **Table S4**.

First, we added the following footnotes to the column names to provide more detail on what these columns represent:

- **Production (1000s mt):** [1] Average annual production of unstudied species in a taxonomic group harvested in vulnerable EEZs from 2014-2023 (FAO, 2024)
- **Number of species harvest:** [2] Number of harvested species in the taxonomic group based on FAO fisheries and aquaculture production data (FAO, 2024)
- **Number of species total:** [3] Number of species in the taxonomic group based on FishBase and SeaLifeBase species databases (Palomares and Pauly, 2025)

Second, we made the "Species" column more specific by renaming it "Species in the top-20 of annual harvest without a depuration rate".

We hope that the revisions to the methods described in the response to the comment three comments below this one also helps to increase clarity on the origin of **Table S4.**

In the text, first of all, line 483, it is 2 and not 3 of the top 5 species that have not been studied.

Thanks for spotting this error. We changed "three" to "two".

Line 486: the assessment of 7 of the 10 species not studied. This statement seems strange to me when compared with the rest of the paragraphs. Should it be the assessment of 7 species not studied that are found in Table S4 out of the 7 not analysed that are found in Figure 5?

We really appreciate this comment because we were also confused when revisiting this text, which indicates how confusingly worded it was!

As a result, we decided to simply delete this category of result from this section. To illustrate, all of these results have been changed from a syntax like this:

> "Thirteen of the top-20 most harvested marine filter-feeding mollusc species from countries exposed to PSTs have had their PST depuration rates studied (**Fig. 6A**), the highest of any of the biotoxin types (**Fig. 6**). However, two of the top-5 species have not had their PST depuration rates examined: constricted tagelus (*Sinonovacula constricta*) and blood cockle (*Tegillarca granosa*). PST depuration rates for 1 new order, 2 new families, and 4 new genera could be gained through the evaluation of 7 of the 10 unstudied species (**Table S4**). In particular, evaluation of blood cockle (*Tegillarca granosa*) would provide information for 8 harvested and 360 total species in the Arcoida order."

To a syntax like this:

> "Thirteen of the top-20 most harvested marine filter-feeding mollusc species from countries exposed to PSTs have had their PST depuration rates studied (**Fig. 6A**), the highest of any of the biotoxin types (**Fig. 6**). However, two of the top-5 species have not had their PST depuration rates examined: constricted tagelus (*Sinonovacula constricta*) and blood cockle (*Tegillarca granosa*). Notably, the evaluation of blood cockle (*Tegillarca granosa*) would provide information for 8 harvested and 360 total species in the Arcoida order (**Table S4**)."

This is much more clear.

Cette structure de paragraphe est répétée pour les différents types de toxines mais reste très peu clair pour le lecteur. Toute cette partie et en particulier comment lire la table S4 et comprendre comment elle a été générée devrait être développée. Comment le total du nombre d'espèces exploitées et total ont été calculés ? Est-ce que ce nombre est mesuré dans la région où on observe les différents syndromes ?

We translated this comment as:

> "This paragraph structure is repeated for the different types of toxins but remains very unclear to the reader. This entire section, and in particular how to read Table S4 and understand how it was generated, should be expanded upon. How were the total number of exploited species and the total number calculated? Is this number measured in the region where the different syndromes are observed?"

We took several actions to address this comment.

First, see how we clarified **Table S4** in the response to the comment three comments above this one.

Second, we added the following underlined text to the methods to clarify how the number of harvested species and the total number of species within each taxonomic group was enumerated:

> "To further illustrate the value of these estimates, we quantified the (1) number of currently harvested species in the taxonomic group using FAO fisheries and aquaculture production data (FAO, 2024) and (2) total number of species in the group using the FishBase and SeaLifeBase species databases (Palomares and Pauly, 2025), which captures potential future harvest diversity. For this illustration, we enumerated all species, even if they do not currently occur in an EEZ exposed to a biotoxin syndrome (**Fig. S4**), given that exposure is likely underestimated and that climate change could change the future distribution of marine HAB and fishery species (Townhill et al., 2018)."

Specifically, the revised text attempts to better convey the source of the information (i.e., the FAO production data and the FishBase/SeaLifeBase species databases) and better explain that it represents diversity in all EEZs, not just those exposed to HABs. It also attempts to explain the rationale for this decision.

Ligne 497 et 498 comment cette information a-t-elle été identifiée ?

We translated this comment as:

> "Lines 497 and 498, how was this information identified?"

Based on the revisions described above, we hope it is now clear that these enumerations are based on the FAO production data and FishBase/SeaLifeBase species database (**Table S4**). As a reminder, this comment referred to this sentence:

> "In particular, evaluation of Manila clam (*Ruditapes philippinarum*), striped venus (*Chamelea gallina*), or northern quahog (*Mercenaria mercenaria*) would provide information for 22 harvested and 428 total species in the Veneridae family."

Ligne 501 à 505 le texte semble dire le contraire de la figure 5C en l'occurrence les analyses n'ont pas été faites pour ces 6 espèces

We translated this comment as:

> "Lines 501 to 505 the text seems to contradict Figure 5C; in this case, the analyses were not carried out for these 6 species."

Thank you for spotting this crucial accidental omission of the word “not”. We’ve added the missing “not” and the text and figure are now properly consistent:

> “Only four of the top-20 most harvested marine filter-feeding mollusc species from countries exposed to DSTs have had their DST depuration rates studied (**Fig. 6C**). Notably, six of the seven most intensely harvested species have **not** had their DST depuration rates examined: constricted tagelus, Pacific oyster (*Magallana gigas*), Japanese scallop (*Mizuhopecten yessoensis*), blood cockle, Chilean mussel, and Atlantic sea scallop (*Placopecten magellanicus*).”

All the data analyses and results produced in paragraph 3.2 are not fully exploited in the rest of the manuscript. The authors should propose a more explicit sampling strategy, or at least present it in the discussion so that readers can fully appreciate the results of this publication, which are very rich but difficult to interpret/understand.

We hope that our efforts to clarify the methods and to simplify and clarify the results (both described in the responses to other comments) have helped to clarify and elevate these results.

The last section of results, 3.3, is also very interesting but very difficult to understand. The first paragraph, lines 556 to 564, needs to be made more explicit, as the various parameters listed have not been presented either in the materials and methods section or in the captions of the figures or tables. Based on the information currently available in the manuscript, I was unable to read/understand/interpret this paragraph.

We are grateful for this comment because not all of the Bayesian performance metrics presented in *section 3.3* (results) or **Table S5** were clearly defined in the methods text (*section 2.3*). We took several actions to correct this.

First, we added the following underlined text to the methods (*section 2.3*) to define the significance of $\hat{R}$, the ΔELPD, and the ΔELPD-SE. This should prepare the reader to be able to interpret the results presented in the first paragraph of section 3.3 and **Table S5**:

> “We fit all four models using a Bayesian estimation approach implemented through the *brms* R package (Bürkner, 2017). <u>We evaluated model convergence by confirming that $\hat{R}$, a statistic that compares the within and between chain variance, was less than 1.01 (i.e., all chains have converged on the same posterior).</u> The models were compared using efficient approximate leave-one-out cross-validation (LOO) using the *loo* R package (Vehtari et al., 2024), where the best model was identified as the one with the largest expected log predictive density (ELPD), which characterizes the ability of the model to predict the data withheld in the cross-validation <u>and thus its ability to make out-of-sample predictions. A model is considered to be significantly better than an alternative</u>

model if the difference in their ELPD's (ΔELPD) exceeds the difference in the standard errors of their ELPDs (ΔELPD-SE). The goodness-of-fit of all four models was also evaluated through the estimation of a Bayesian R^2 (Gelman et al., 2019)."

We also improved **Table S5** by removing two unnecessary metrics (LOOIC and P, which were previously undefined) and adding a key missing metric (ΔELPD-SE):

Table S5. Performance statistics used in model selection.

Model	ELPD (SE)	ΔELPD	ΔELPD-SE	R^2 (CI)
Taxonomically nested random effects	-229.4	0	0	0.67 (0.6-0.72)
Species random effects	-229.7	-0.3	1.1	0.66 (0.59-0.71)
Phylogenetic random effects	-230.8	-1.3	1.7	0.65 (0.57-0.71)
Fixed effects only	-255.4	-26	6.7	0.49 (0.4-0.56)

** ELPD = expected log predictive density (higher = better); R^2 = Bayesian R^2 (higher = better); SE = standard error; CI = 95% confidence interval*

Finally, we added the following underlined text to clarify the presentation of these metrics in the results (*section 3.3*):

"The models with taxonomic random effects exhibited similar performance to each other as the ΔELPD-SEs exceeded the ΔELPDs. However, based on the ELPD point estimate, the greatest statistical support was for the taxonomically nested random effects model (ELPD=-229.4, R^2=0.67) followed by the species random effects model (ΔELPD=0.3, ΔR^2=0.01) then the phylogenetic random effects model (ΔELPD=1.3, ΔR^2=0.02). The taxonomically nested random effects model was therefore selected as the best performing model."

Furthermore, the names of the different models do not correspond to the exact names of the models in the materials and methods section or the exact names of the models in Table S5, which makes this paragraph even more difficult to read. This information needs to be clarified.

Table S5: is it possible to use the same model names as defined in the materials and methods and perhaps list them in the same order, either in the materials and methods or in the results?

We revised the text, tables, and figures to consistently describe the four models as the:

1. "Fixed effects only" model
2. "Phylogenetic random effects" model
3. "Taxonomically nested random effects" model
4. "Species random effects" model

We appreciate you flagging this inconsistency, as this would have caused confusion.

The various parameters measured have not been explained in the materials and methods, which could improve the readability and comprehension of this part of the manuscript, which remains quite difficult to follow.

> Please see our response to the comment related to *Section 3.3* Lines 556-564 above where we explain how we added text to the methods to introduce these parameters.

In addition, Figure S9 is very difficult to read. What do the different models numbered from 1 to 10 represent? What do they refer to? This figure and its elaboration need to be clarified.

> This figure shows the posterior predictive check plots that are used to examine the fit of Bayesian models to data. We added the following underlined text to the caption to increase clarity for non-Bayesian readers while also retaining Bayesian reporting conventions for Bayesian modelers (i.e., y, y_{rep}, draws):
>
> > "**Figure S9.** Diagnostic plots for the taxonomically nested random effects model (the best performing model) showing **(A)** the distribution of the residuals in 10 posterior draws; **(B)** distribution of the response variable (log depuration rate) in the data (y) versus 10 posterior draws (y_{rep}); and **(C)** correlation between the response variable in the data (y) versus 10 posterior draws (y_{rep}). In **(C)**, the diagonal line represents the one-to-one line and each facet represents a set of posterior draws. The model was judged to pass these posterior predictive checks because the (A) residuals are normally distributed around zero; (B) the shape of the distributions of predicted depuration rates (y_{rep}) match the shape of the distribution of observed depuration rates (y); and (C) the predicted depuration rates (y_{rep}) are correlated to the observed depuration rates (y) without concerning patterns of bias."

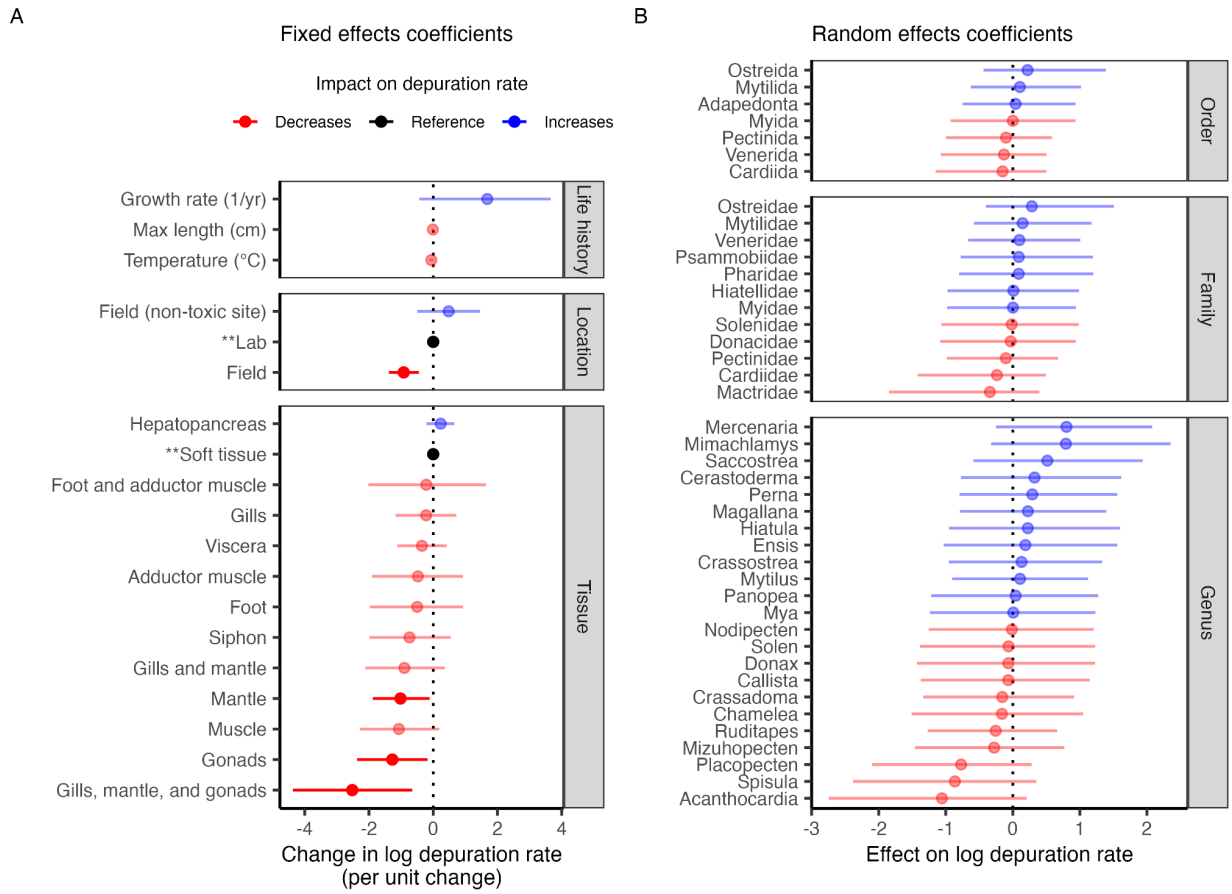
The reason for choosing one model over another is also not clear in the discussion and should be made clearer.

> Please see our response to the comment related to *Section 3.3* Lines 556-564 above where we explain how we added text to the methods and results to better describe how various Bayesian goodness-of-fit statistics were used to select the best model.

Figure 6: this figure should be explained more clearly both in the text (M&M and results) and in the caption. In particular, the difference between the two models: what exactly are they and how should this figure be read? The reference to * is unclear. How were these values chosen?

> We took several actions to address this comment.
>
> First, we changed the titles in the figure to read "Fixed effect coefficients" and "Random effects coefficients" to clarify that these are coefficients from the same mixed effects

model (the taxonomically nested random effects model, which was the best performing model). They are not coefficients from different models.



Second, we edited the caption to make it more clear that these are coefficients from the same model. The underlined text used to read "final model":

> "**Figure 7.** The **(A)** fixed effects and **(B)** random effects coefficients estimated by the taxonomically nested random effects model (the best performing model)."

In regression models with categorical covariates, coefficients are estimated relative to a reference level. We opted not to explain this as this is a widely known statistical concept and we think it is sufficiently explained by the following text in the caption:

> "The categorical fixed effects (location and tissue) are estimated relative to a reference level, which is marked with asterisks (**) and is indicated in black."

Finally, we hope this clear things up sufficiently and that we do not have to add any new text to the methods or results. We feel that the current text describing this figure in the methods and results is sufficient:

Methods: “We evaluated the coefficients and conditional effects of predictors in the best fitting model to understand the impact of each variable on PST depuration rates.”

Results: “The taxonomically nested random effects model estimated that depuration rates are significantly slower (smaller) in the field relative to the laboratory (**Fig. 7A; Fig. S10A**). On average, field depuration rates are 2.5 times slower than laboratory depuration rates. The model estimated non-significant impacts of the three life history variables, which exhibited a mixture of non-significant positive (growth rate) and negative (maximum length, preferred temperature) effects. The model estimated significant differences in PST depuration rates among bivalve tissues with the hepatopancreas and soft tissues exhibiting faster depuration rates than other tissues. The conditional effects of the fixed effects variable, which estimate the depuration rate with all other effects estimated at their average, are shown in **Fig. S10**. The greater statistical support for this model over the “fixed effects only” model indicates a significant impact of taxonomy on depuration rates with the relative impacts of taxonomy on depuration rates illustrated in **Fig. 7B**. The greater support of this model over the phylogenetic random effects model suggests that, based on the current data, while depuration rates vary by clade, they do not vary based on the relatedness of clades (**Table S5**).”

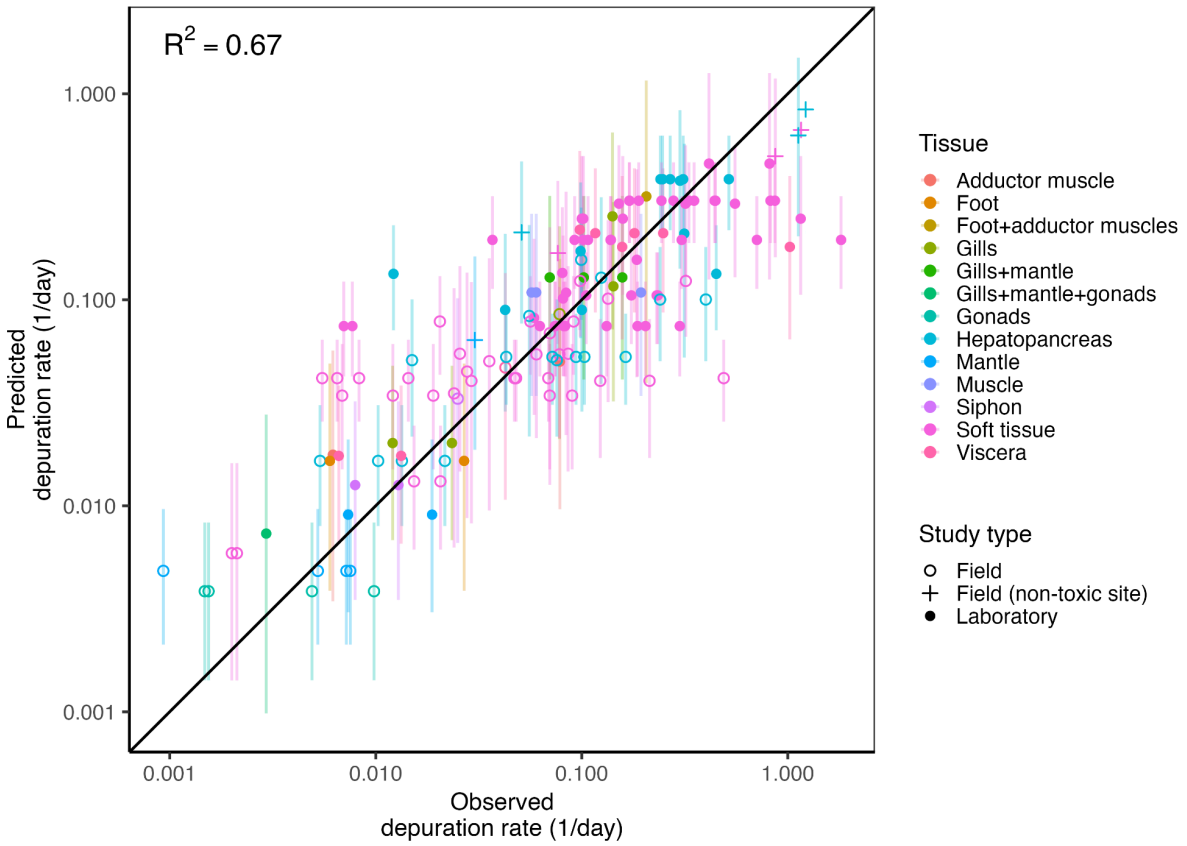
All of these analyses make it possible to generate Table S6 and Table S7 on the predicted depuration for different toxins for PSTs, which is very comprehensive and promises to be useful to the profession. But what about the reality and the species for which there are actual depuration rates? Are these predictions close to reality?

In these two tables, it would be useful to add the species for which depuration data are available and to add a column showing the observed depuration rates for the species for which this rate has been observed. These additions would thus reassure the reader of the accuracy of the models for predicting PST toxin depuration rates.

Figure 7: In relation to the comment on Tables S6 and S7, is it possible to have both the estimated values and the measured values, where they exist, in order to compare the predictive model data with the actual data? This information would reinforce the relevance of this work and this paper.

We address these three comments together as they are all related to the request to better show whether the predictions are correlated to the observations documented in the training database (i.e., the goodness-of-fit of the model).

We added a new supplemental figure **(Figure S8**) to illustrate the goodness-of-fit as a scatterplot showing the observed versus the predicted PST depuration rates for bivalves.



Please see our response to the related comment in the major comments for details on how we also expanded the caption of **Figure S9** to better explain how these Bayesian posterior predictive check plots also illustrate the goodness-of-fit of the model.

We opted not to add the observed values to **Tables S6** or **S7** because (1) the new scatterplot conveys this information and (2) there are often multiple observations informing each prediction, which would lead to complicated table formatting (e.g., four different studies might examine PST depuration from blue mussel soft tissues in the lab).

Could the same type of table (Fig. 7, Tables S6 and S7) be generated for other types of toxins? In particular, amnesic toxins, for which a certain amount of purification data is already available in the literature. If so, this could be interesting; if not, could the authors explain how much depuration data is needed to make this kind of prediction and table possible? This point should also be addressed more clearly in the discussion, as it is clearly something operational that is expected when reading the introduction to the manuscript.

Please see our response to the related major comment on why we were only able to model/predict PSTs in bivalves.

Indeed, in line 636 of the discussion, the authors specify that managers should assume that depuration is 2.5 times slower in the field than in the laboratory. These results are true for PSTs, but is this true for other toxins?

Thanks for catching this. Indeed, this value has only been quantified for PSTs and not for other toxins. We added the following underlined text to clarify this point:

> "For example, managers could assume that depuration occurs 2.5 times more slowly in the field, the average estimated <u>for PSTs</u> by our best performing Bayesian regression model."

To help managers make judgements for the magnitude of this adjustment factor for other toxins and clades of species, we added the observed adjustment factor to **Figure 5**.

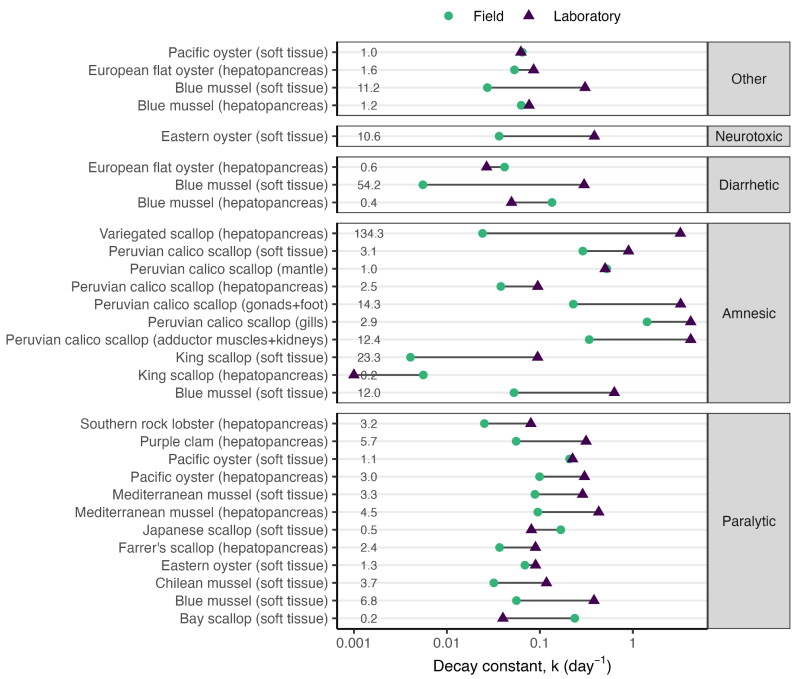


Figure 5. *A comparison of average depuration rates in the field versus the laboratory by species, tissue, and toxin. <u>The number on the left indicates how many times slower depuration is in the field than in the laboratory (values less than 1.0 indicate the rare cases in which depuration was estimated to be slower in the laboratory than in the field).</u>*

Figure 8: line 1216 "species are represented in blue", but there does not appear to be any blue in this figure; this information needs to be changed.

Thank you for catching this error. We changed "blue" to "orange".

Discussion

As previously stated in line 636, it is unclear whether the authors are referring to PSTs or all toxins. This paragraph could be expanded upon slightly.

Thanks for catching this. Indeed, this value has only been quantified for PSTs and not for other toxins. We added the following underlined text to clarify this point:

> "For example, managers could assume that depuration occurs 2.5 times more slowly in the field, the average estimated for PSTs by our best performing Bayesian regression model."

See our response to two comments above this one about the adjustment to **Figure 5** that helps to provide managers with information on other toxins and clades.

Line 652: The authors suggest that managers could use this information and knowledge of parameters such as temperature, food availability and condition index to recalculate the optimisation of toxin monitoring data. Could you elaborate on this point and explain how managers can now use all the data generated to achieve this objective?

We took two actions to address this comment.

First, we added text to clarify that these upwards and downwards adjustments will need to be made based on their expert judgement.

Second, we developed a web tool to support managers with the calculations required to forecast depuration timelines and schedule optimized biotoxin monitoring.

These adjustments are reflected in the following underlined text:

> "Then, managers can use knowledge of recent temperature, food availability, or body condition to adjust the expected depuration rate based on their expert judgement and recalculate the optimized cadence of biotoxin monitoring. To assist managers with such calculations, we developed a webtool to support forecasting depuration timelines and optimizing biotoxin monitoring here: https://emlab-ucsb.shinyapps.io/depuration_forecaster/"

Lines 660 to 662: Is it possible to elaborate on this sentence and explain in particular how to develop such a strategy? Are there any examples in the literature and what did they do?

We added the following underlined citation and text to exemplify how farmers might achieve this:

> "Another strategy could be to limit or inhibit feeding during a bloom to avoid biotoxin accumulation (Mukai et al., 2026) but also to reduce body condition and increase hunger to stimulate rapid consumption and depuration after the bloom abates (Castberg et al., 2004). For unfed aquaculture, this could be achieved by increasing stocking density, moving bags to less productive areas, or sinking bags to less productive depths."

Mukai, K., Yano, R., Kitatsuji, S., Hotta, T., Yamazaki, H., Takai, Y., Hano, T., Nishimaki, T., Shikata, T., 2026. Feeding Cessation Improves Red Tide Resistance in Aquacultured Yellowtail. Aquac. Res. 2026, 6331079. https://doi.org/10.1155/are/6331079

Line 707: the authors mention that it would be interesting to take the same approach for toxins other than PSTs. Why has this not been done? Is it possible to do so? If not, please explain why you have not done so for other toxins.

Please see our response to this comment in the “Major Comments” section at the top of the review. Briefly, it would be extremely valuable to do this work for other toxins, but there is currently insufficient data to support a model with good predictive skill.

Lines 722 to 726: the authors say that they started from the quantification of 45 species-toxin combinations in 30 studies. I am not sure I understand this point. This work is not clearly visible in this publication. Unless it is a recommendation. This sentence needs to be clarified/explained.

In this paragraph, we are trying to make the point that valuable information on biotoxin depuration rates exists within already collected monitoring data that just needs to be analyzed. We then try to highlight that our study has uncovered depuration rates from many monitoring studies, but there are probably many more in existence, since these studies wouldn’t have been discovered by the search terms used in our systematic literature review. We added the following underlined text to clarify this point:

> “<u>Our study has begun the process of uncovering depuration rates from already collected monitoring data: specifically,</u> we quantified depuration rates for 45 species-toxin combinations from 30 field studies that did not directly estimate depuration rates but were discovered in our systematic review due to the inclusion of depuration-related words in their abstract or keywords (e.g., (Haya et al., 2003; Kvrgić et al., 2022; Rourke et al., 2021)). However, this still overlooks the likely large number of biotoxin monitoring programs that are either not documented, are documented without these keywords, or are only documented in the grey or non-English literature.”